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# Machine Learning–Driven Temperature Prediction for Smart-City HVAC Optimization Using IoT Sensor Data

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***Abstract*—In modern smart-city infrastructures, accurate prediction of temperature has become a critical functional demand that enables pre-emptive HVAC control, energy efficiency, and occupant comfort. This study develops and validates a suite of ML regression models to temperature prediction using an IoT dataset comprising 405,184 records collected at regular intervals at seven days which contains multiple environmental variables—temperature, motion, CO, LPG, smoke, light, and humidity. Exploratory data analysis constructed for testing temporal variation and inter-variable correlations. Under a chronological 80/20 split to avoid temporal leakage, XGBoost achieves robust performance, with $R^2 = 0.934$, $\mathrm{RMSE} = 0.334$, and $\mathrm{MAE} = 0.304$. For interpretability, A detailed diagnostic study (including residual analyses, error-frequency distribution, and condition-aware evaluation) shows most predictions within ±1°C. Reliability of the model is strengthened through feature importance & SHAP-based approach that highlights humidity, light, and gas sensors as dominant predictors in the temperature dynamics. Paired t-tests ensure that ensemble methods significantly outperform baseline with linear and single-tree. This study is transparent and fast, provides fine-grained forecasts and supports comfort-driven HVAC scheduling by early detection of thermal fluctuation.**

***Index Terms*—Internet of Things (IoT), Machine Learning, Temperature Prediction, Sensor,XGBoost, Random Forest, SHAP.**

## I. Introduction

Reliable environmental monitoring has become a crucial demand in the context of climate variability. The rapid expansion of IoT-based technology now enables continuous, fine-grained collection across diverse indoor settings of environmental sensor data. Among these various operations with IoT sensor data, temperature measurement is particularly significant for its impact on energy use, environmental sustainability, industrial processes, and occupant comfort assurance. Therefore for efficient HVAC operation in smart buildings, indoor-air-quality systems, and healthcare or resource-management facilities accurate short-term temperature prediction is crucial. For nonlinear and dynamic nature of indoor microclimate sensor data, traditional forecasting methods often struggles. However, in this situation ML-based approaches can leverage large heterogeneous sensor streams and the dataset's dynamic features to capture these complex patterns and variations. The present approach is conducted motivated by this capability and evaluates several ML based models using an IoT sensor data collected from multiple identical Raspberry Pi device for short-horizon temperature forecasting . These sensor devices produce a comprehensive multivariate dataset by capturing CO concentration, relative humidity, lighting state, LPG levels, smoke density, and motion activity which is a significant need for data-driven temperature prediction. The main contributions of this work are summarized below:

- We illustrate a comparative analysis with the performance of a range of ML models for accurate temperature prediction using real-world dynamical IoT sensor data.
- We conduct exploratory analysis with visualizing corelation among data features to understand temporal behavior and inter-variable relationship within data.
- Beyond model comparison, We perform a detailed error study(residual plot, error histogram patterns and error-frequency distributions) along with statistical check(paired $t$-test) to evaluate the consistency of models performance.
- We visualize feature importance plot and applied SHAP-based explanations, providing insight into how different sensor readings influence temperature predictions.

This paper is organized as: Prior work discussed in Section 2, Section 3 narrates the dataset and preprocessing techniques. Section 4 represents the methodology. In Section 5 discussed about findings & conclusion & future work in Section 6 .

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## II. Related work

Our study was motivated by the need for efficient, generalizable temperature-prediction models that can operate reliably in real-time IoT environments. Although considerable progress has been made, temperature-related IoT data vary widely across domains, leading to diverse modeling approaches and mixed performance.In agriculture, deep-learning models trained on sensor networks in southeastern Spain achieved high accuracy (e.g., $R^2 \approx 0.95$) [1], [2]. Sensor-based monitoring has also been applied in biomedical laboratories, where refrigeration data supported anomaly detection through dynamic thresholding [3], [4]. Meteorological studies from Kuala Terengganu found that ANN models outperform traditional predictors for temperature and humidity [5], while ARIMA has shown advantages over linear regression in both indoor and outdoor settings [6].Cloud-integrated IoT pipelines using MQTT and AWS have likewise been explored for short-term forecasting [7], and several works highlight the role of land-surface temperature, elevation, and Julian day in improving prediction accuracy [8]. Additional studies demonstrate how sensor-based machine-learning systems support applications such as smart-door monitoring [9] and multi-room temperature estimation using compact neural networks [10]. High-resolution sensing has further advanced the field: DTS systems have achieved fine-grained accuracy as low as $0.13\,^{\circ}\mathrm{C}$ in field trials [11], while calibration and shielding have been shown to significantly reduce sensor bias [12]. Environmental applications include identifying temperature-safe zones in aquatic ecosystems using DTS and thermal imaging [13], and estimating indoor temperatures with minimal sensor requirements [14]. Vertical DTS setups have also demonstrated very high precision in stratified environments [15]. Overall, prior research highlights substantial progress but also reveals gaps: many systems rely on specialized sensors, domain-specific setups, or high-cost instrumentation. There remains limited work on lightweight, interpretable machine-learning baselines built directly on accessible IoT sensor streams for real-time environmental monitoring. Our research addresses this gap by conducting practically best-fitted models for smart-building temperature control and short-horizon microclimate prediction.

## III. Methodology

Initially, our main plan was to construct a workflow on short-horizon temperature forecasting using ML based techniques. Fig.1 illustrates the overall steps of our approach, representing the scenario of data collection, preprocessing steps, and analysis of overall model.

### A. Data

We constructed our work with a publicly available dataset named *Environmental Sensor Telemetry dataset [16]* that collected from different environment conditions. The features of the dataset including temperature, humidity, CO, LPG, smoke, light, and motion. The dataset has total 405,184rows timestamped records which is collected over a seven-day period from July 12,2020 to July 19, 2020. The sensor data measurement was collected with three IoT sensor each identified by a unique device ID as named *00:0f:00:70:91:0a, 1c:bf:ce:15:ec:4d, & b8:27:eb:bf:9d:51*. These multivariate signals of environmental IoT sensor generate a descriptive view of indoor microclimate dynamics.

*1) Data Behaviors and Feature Relationships*

The data analysis is designed for finding and understanding variable behaviors and interactions between them. Visual representation of temporal trends like temperature and humidity over time and analysing pairwise relationships like CO vs. LPG or smoke illustrate a regularized patterns across sensors. The correlation matrix as represented in Fig. 2 indicates a moderate negative correlation between humidity and temperature ($r = -0.41$), in the contrary CO, LPG, and smoke generate weak positive correlations. These revealed patterns proves the nonlinear and multivariate nature of the collected data.

*2) Data Preprocessing*

Before fed the dataset into models for training. we need to prepare data for best outcome. Firstly, we applied standardization technique for transforming each numerical feature for having a mean of zero and a standard deviation of one. With maintaining a consistent preprocessing pipeline that supports fair comparison across all algorithms are used in this approach although the tree-based methods are scale-invariant largely. Standardization method also stabilize gradient-based optimizers, accelerates convergence, and improves generalization for making models sensitive to feature scaling such as $k$-NN, and GBoost method.

### B. Machine Learning Models

For forecasting indoor temperature from IoT-based environmental signals we applied six different regression model to identify how model works and response. After the selection the relevant features that are responsible for predicting temperature (CO concentration, humidity, lighting state, LPG, motion, and smoke), We standardize continuous features(binary features (light, motion) are encoded as 0,1) using statistics computed on the training set only. We adopt a chronological split to avoid temporal leakage as the earliest 80% of data features for training and the latest 20% for testing. Exclusively on the training data and applied to the test set without re-fitting, all model selection and scaling steps are fit.
Linear Regression (LR), Decision Tree Regression (DT), $k$-Nearest Neighbors Regression (k-NN), Random Forest Regression (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost) in total six regression models were trained and evaluated: After performance evaluation, we observed that XGBoost consistently delivered the strongest performance on predicting, proving its capability for handling heterogeneous and dynamical character variety of sensor data. In this context XGBoost's effectiveness can be caused for its several characteristics as mentioned below:

- **Tree Ensembled Boosting:** The gradient-boosting framework has additional capability to capture nonlinear rela-

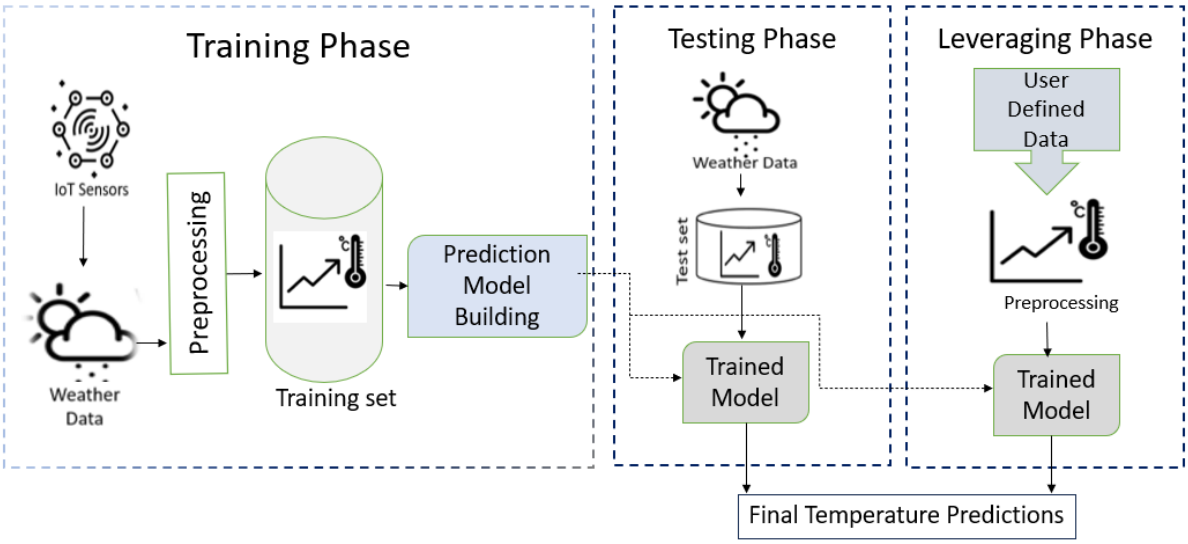


Fig. 1: Temperature Prediction Process

Fig. 2: Correlation heatmap of environmental Features

tionships and complex interactions by its tree ensembling character.

- **Regularization:** Improving generalization to unseen time periods and sensor conditions, its Built-in L1 and L2 penalties character help reduce overfitting.
- **Missing or Noisy Data Handling:** XGBoost is robust to real-world data imperfections, making it well-suited and resilient for where signal interruptions or noise can occur in environmental IoT data .
- **Reliability and Scalability:** For speed and resource efficiency, enabling fast training even on large datasets, its algorithm structure is optimized explicitly.

## IV. Result & Discussion

The primary focus of this framework is to predict temperature using the available features collected from environmental IoT Sensor. Each ML based model was validated using predefined evaluation protocol to confirm a fair and transparent comparative analysis. The whole assessment clarified that the models perform robustly on this sensor data, with the detailed performance evaluation is represented in Table I

TABLE I: Regression model performance for short-term temperature prediction

| Model | $R^2$ | RMSE | MAE | Inf. (ms) |
|---|---|---|---|---|
| RF | 0.933 | 0.701 | 0.306 | 0.055 |
| k-NN | 0.929 | 0.718 | 0.315 | 0.109 |
| DT | 0.925 | 0.737 | 0.314 | 0.001 |
| XGB | 0.934 | 0.334 | 0.304 | 0.005 |
| GB | 0.899 | 0.857 | 0.491 | 0.002 |
| LR | 0.781 | 1.262 | 0.727 | 0.000045 |

where all models has sub-millisecond per-sample inference time, proving suitability for real-time HVAC deployment. Additional validation analyses were performed to evaluate robustness and practical significance beyond point estimates. Bootstrapped confidence intervals indicate stable predictive performance, achieving an $R^2$ of 0.919 (95% CI: 0.911–0.927) and an RMSE of 0.768 (95% CI: 0.728–0.809). Furthermore, with $R^2$ scores ranging from 0.44 to 0.99 and a mean $R^2$ of 0.735 rolling-window validation across multiple temporal splits. These consistent performance demonstrate robustness and reliability against temporal variation and reduce dependent split bias risk.

### A. Data Analysis

In our approach, we conduct exploratory data analysis and find-out consistent patterns when feature are interacting internally and there was stable correlations across features, it indicate that the data was well organized and that the assessment of models' performance is both reliable and robust. As showed in Fig. 3, the first plot highlight the diversity of environmental conditions in insights of data features and illustrate the temporal temperature variation across different sensors. The next plot represents the corelation between

temperature and humidity, revealing a transparent and clear-insights association between the two features.

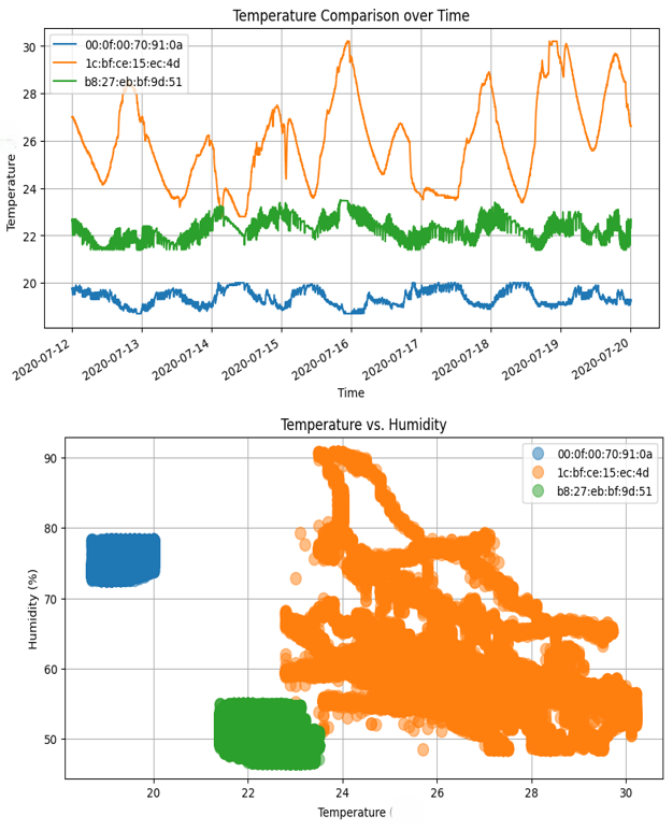


Fig. 3: Data Analysis

### B. Testing Phase

In this phase, The performance of all models are evaluated with test data. Fig.5 illustrates the comparison between actual and predicted temperature values for the test set. A residual

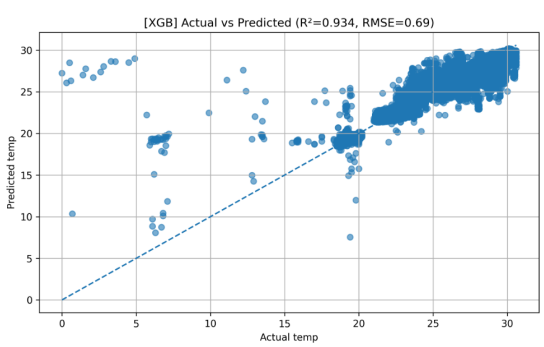


Fig. 4: Actual vs predicted temperature

analysis was performed to further investigate error behavior. Fig.5 shows the residual plot, where residuals are distributed around zero across different temperature ranges. Additionally, the distribution of prediction errors is presented in Fig.6. The histogram confirms that most predictions fall within a

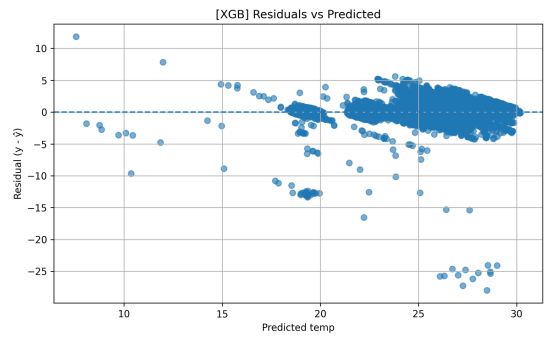


Fig. 5: Residual Plot of prediction errors

small error margin (e.g., ±1°C). Finally,statistical significance

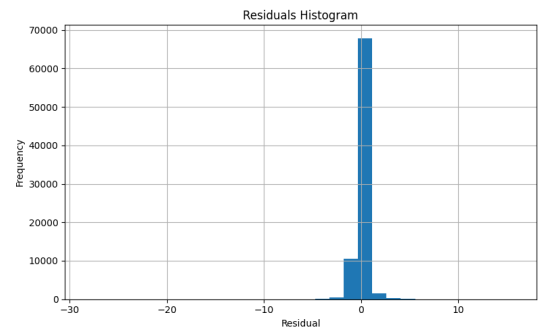


Fig. 6: Error Distribution Histogram

tests were conducted to validate the robustness of model comparisons that proves over baseline models are statistically significant.

### C. Comparative Analysis

To ensure a fair comparison, all models were trained and tested using an identical preprocessing, data split, and validation technique. Comparative bar chart in Fig.7 showing performance of six regression models (LR, DT, k-NN, RF, GB, XGBoost) based on MAE, MSE, and R². Feature importance

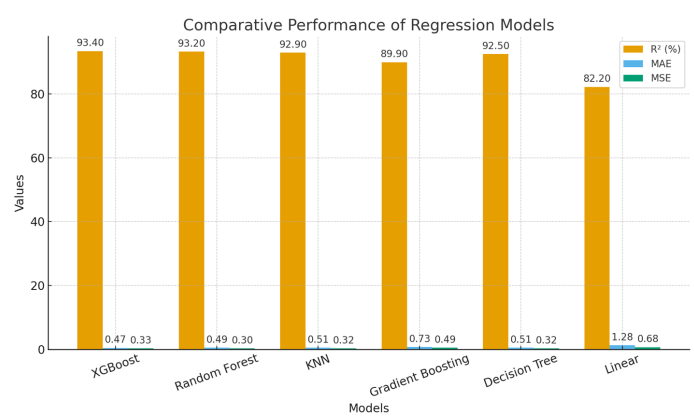


Fig. 7: Comparative Analysis

ranking from the XGBoost model, highlighting the most

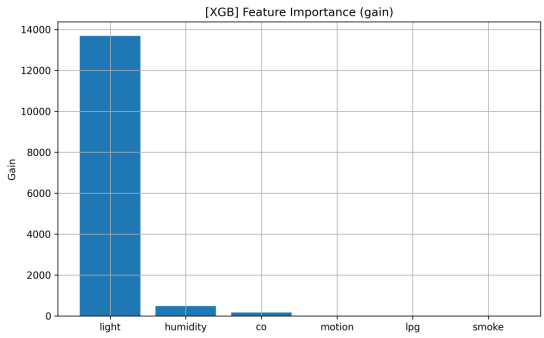


Fig. 8: Feature Importance From XGBoost

influential predictors of temperature variation in Fig.8: SHAP summary plot in Fig.9 illustrating the contribution and direction of each feature towards XGBoost model predictions where Light intensity captures solar radiation effects and indoor heat gain, while CO concentration acts as a proxy for human occupancy and associated metabolic heat. Error and residual

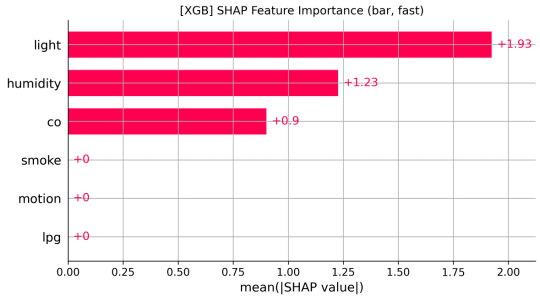


Fig. 9: SHAP summary plot

analysis revealed that ensemble methods produced smaller and more stable deviations compared to baseline models. SHAP and feature importance further clarified that variables such as light,humidity and CO strongly influenced predictions. Table II presents the results of the paired $t$-tests on model errors. Most comparisons are statistically significant ($p < 0.05$), showing that ensemble models—RF, Gradient Boosting, and XGBoost—consistently outperform baseline methods such as LR, DT, and k-NN. There are only one non-significant result occured between DT and k-NN, that represents these two models generate similar error behavior under the assessment protocol.

### D. Leveraging Phase

The short summary of our proposed sframework follows a straightforward steps:

IoT Data → Preprocessing → ML Model → Pred. Temp.

For proving the usability of the trained models, we analyze their behavior and response using a representative user-defined input. The incorporation of such additional inputs make a different assessment of our model for testing its reliability and interpretability with real-world data beyond the training data.

TABLE II: Pairwise paired t-test between models' absolute errors on the same test instances ($p < 0.05$ indicates significant difference).

| Model A | Model B | p-value | Significance |
|---|---|---|---|
| LR | DT | <1e-300 | Significant |
| LR | k-NN | <1e-300 | Significant |
| LR | RF | <1e-300 | Significant |
| LR | GBoost | <1e-300 | Significant |
| LR | XGBoost | <1e-300 | Significant |
| DT | GBoost | <1e-300 | Significant |
| GBoost | XGBoost | <1e-300 | Significant |
| k-NN | GBoost | <1e-300 | Significant |
| RF | XGBoost | <1e-300 | Significant |
| RF | GBoost | <1e-300 | Significant |
| k-NN | XGBoost | 3.208e-203 | Significant |
| DT | XGBoost | 5.723e-167 | Significant |
| DT | RF | 1.16e-38 | Significant |
| k-NN | RF | 8.172e-25 | Significant |
| DT | k-NN | 0.1476 | Not Significant |

TABLE III: Temperature Prediction from Different Regression Models

| **Model** | **Temperature (°C)** |
|---|---|
| XGBoost Regression | 25.7 |
| Random Forest Regression | 26.4 |
| k-NN Regression | 27.4 |
| Gradient Boosting Regression | 26.2 |
| Decision Tree Regression | 27.4 |
| Linear Regression | 26.4 |

For this evaluation, as input the mentioned data is applied: device ID: *b8:27:eb: bf:9d:51*; CO concentration was 0.004; humidity was in 55.0% ilevel; light was detected(*True*; LPG was 0.006 with no detected motion(*False*; and smoke was 0.018 level. The same user input data was fed into all pre-trained models to observe how their respective prediction outcome differ under identical environmental scenarios. The TABLE III illustrate that there are a little variance prediction ($\pm 1$ degree) for each models that confirms as models are stable with there outcomes even in various methods.

### E. Discussion

Most model comparisons are statistically significant ($p < 0.05$) as the paired $t$-test results showed, verifying that ensemble methods—RF, GBoost, and XGBoost consistently outperform baseline models. Error histogram, Residual analysis delivered small errors, with larger deviations only at extreme temperature ranges, which is flexible for real-world environmental data. Light, humidity, and CO is the most influential predictors that was found from feature-importance results and SHAP explanations and this findings also reinforce the interpretability and robustness of the models. DNN approaches remain promising for capturing longer-term temporal patterns

TABLE IV: Qualitative Comparison with Existing Studies

| Study | Dataset | Method | Main Focus |
|---|---|---|---|
| NYC [1], [2] | GeoTab Data (USA) | LSTM Neural Network | Air temperature forecasting with missing-data interpolation |
| Spain [2] | Cieza & Moratalla (Spain) | Deep Learning Model | Crop-oriented temperature prediction using IoT sensors |
| Bio-Lab [3], [4] | 100 Biomedical Sensors | Dynamic Threshold (Unsupervised) | Anomaly detection in refrigeration environments |
| Malaysia [5], [6] | Meteorological Data | ANN / ARIMA / LR | Daily and monthly temperature & humidity prediction |
| AWS [7] | Cloud MQTT Stream | Linear Regression | Cloud-based environmental data integration and prediction |
| Indoor [8], [10], [11] | Smart Building Sensors | ANN / 4-Neuron Network | Multi-room temperature tracking and sensor calibration |
| DTS [14], [15], [17]–[21] | Ecosystem / Groundwater Data | Distributed Temperature Sensing (DTS) | High-resolution environmental monitoring |
| **Our Work** | Multi-sensor IoT Dataset | **XGBoost Regression** | Real-time, generalized, and efficient temperature prediction framework |

as they were not examined here. At last, the convincing outcome and transparency of the models functionality highlight their efficiency for smart-city construction, enabling adaptive HVAC control, early detection of temperature imbalance, and ensure efficient energy management in modern urban nature. As reflected in Table IV, in earlier research these examination are consistent with trends. It shows that within sensor-driven smart-city environments, Our ensemble-based framework shows stronger flexibility, improved scalability, and greater robustness for real-time learning comparing with many domain-specific temperature prediction models

## V. Conclusion & Future Work

This research demonstrate a comparative exploration of six ML based models using a dynamic IoT sensor dataset containing a huge environmental observations. The evaluation proved that ensemble methods—like XGBoost—are highly capable at capturing complex relationships among environmental features and generate stable, realistic temperature prediction. Even under changing of environmental conditions, error showed a little variation ($\pm 1°\mathrm{C}$) that enables more accurate HVAC setpoint control, reducing overcooling and overheating events and improving thermal comfort, with potential energy efficiency advantage in smart-city environments.

The scholarly addition of this work lies in its systematic analysis of various regression models, providing insights into the importance of characteristics, through paired $t$-tests statistical validation also validated, and interpretability is also checked using SHAP. Finally, these empirical assessment provide a transparent and strong baseline that suitable for smart-building and smart-city infrastructure applications.

**Limitation:** Although our work ensured generalization, the dataset spans seven days that suited for short-term temperature forecasting. Long-term seasonal modeling is beyond the scope of the research and will be addressed in future work using extended dataset with DNN time series model architectures.

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